

Technical Framework: Entropy-Driven Feature Aggregation for Industrial Sensor Anomaly Detection

The transition from traditional signal processing to information-theoretic graph models is a strategic necessity for high-reliability Cyber-Physical Systems (CPS). In complex industrial sensor networks—ranging from NASA's Mars Science Laboratory (MSL) to critical water treatment infrastructures—the primary challenge is not merely monitoring individual time-series, but capturing the "implicit relationships" and logical coupling between nodes. Traditional methods such as Generalized Cross-Correlation (GCC) rely heavily on Gaussian assumptions; however, real-world industrial noise and speech-like telemetry are frequently non-Gaussian. By leveraging a "minimum entropy" approach, we can quantify system stability more robustly than second-order statistics. Specifically, maximizing the Multichannel Cross-Correlation Coefficient (MCCC) is equivalent to minimizing joint entropy under Gaussian assumptions, but the entropy-centric model maintains its discriminative power in the non-Gaussian environments typical of failing machinery or interference-heavy telemetry. To quantify the uncertainty and complexity of stochastic processes in a sensor network, we define:

- **Shannon Entropy Rate:** Measures the average information gain per sample for discrete-valued processes, essential for quantifying randomness in digital signal sources.
- **Differential Entropy Rate:** Extends entropy to continuous-valued, discrete-time stochastic processes, providing a complexity measure for analog signals generated by physical sensors.
- **Minimum Joint Entropy:** Serves as a more robust indicator of dependency than standard cross-correlation, particularly when the system is subjected to non-Gaussian noise. These measures provide the mathematical engine required to determine node-level importance within a formal graph structure.

1. Methodological Framework for Sensor Graph Generation

Modern monitoring must evolve beyond treating sensors as isolated temporal streams. We model the system as a directed graph $G = (V, E)$, where structural modeling captures the logical coupling—such as the relationship between a pump pressure sensor and a downstream flow meter—that simple temporal analysis fails to detect. In a CPS context, sensors are rarely physically wired in a mesh, yet their functional interdependencies define the system's "logical architecture." The generation of the sensor graph follows a rigorous procedural sequence:

1. **Node Representation:** Each sensor is represented as a node v_i using learnable vectors in a feature matrix $V \in \mathbb{R}^{n \times d}$, optimized to reflect specific sensor characteristics.
2. **Calculation of Edge Weights:** The influence of node v_j on node v_i is determined via a normalization function: $W_{ji} = \frac{\langle v_j, v_i \rangle}{\|v_j\| \cdot \|v_i\|}$
3. **Directed K-Nearest Neighbor Selection:** To mitigate the "firehose problem" of excessive signal noise and reduce computational overhead, we reject fully connected graph assumptions. Instead, we select the **Top K** related nodes for each sensor based on the highest W_{ji} scores. This directed, K-nearest neighbor structure

improves efficiency by focusing aggregation on high-fidelity logical interactions. Once this structure is established, we can weight the information flow based on its topological importance.

2. Structural Entropy Centrality: The Core Weighting Mechanism

Structural Entropy Centrality defines a sensor's contribution to the system's overall information state. In macroscopic industrial systems, the Second Law of Thermodynamics dictates that entropy is naturally difficult to decrease; therefore, the contribution of a specific node to the system's total entropy is a highly stable indicator of its topological influence. We calculate Entropy Centrality $H(v_i)$ as the delta between the total system entropy and the entropy of the graph excluding that specific node: $H(v_i) = H(G) - H(G \setminus v_i)$

The following structural metrics define the probability of occurrence ($p(v_i)$) within the network:

Metric	Formula / Relation	Description
Node Function	$f(v_i) = \sum_{(v_i, v_j) \in E} \text{weight}(v_i, v_j)$	The sum of weights of directed edges connected to node v_i .
Node Probability	$p(v_i) = \frac{f(v_i)}{\sum f(v_j)}$	The relative structural importance of node v_i within the graph G .
System Entropy	$H(G) = -\sum p(v_i) \log p(v_i)$	The total uncertainty or information density of the network structure.

This "Entropy-Centric Attention" prioritizes nodes based on their **Entropy Decay Strength**. Unlike uniform attention mechanisms, this approach prioritizes nodes that exert the most influence on the system's uncertainty state, ensuring that the detection framework is hyper-sensitive to deviations in critical, logically-central components.

3. GNN Architecture and Feature Aggregation Logic

The structural entropy centrality is integrated into a Graph Attention Network (GAT) framework to process multivariate time-series data. This allows the model to adaptively adjust weights based on the information-theoretic importance of neighboring nodes. The attention coefficient $\alpha_{i,j}$ incorporates the entropy difference as follows: $\alpha_{i,j} = \exp(\text{LeakyReLU}(H(G) - H(G \setminus v_j))) / \sum \exp(\text{LeakyReLU}(H(G) - H(G \setminus v_r)))$

Key architectural differentiators include:

- **LeakyReLU Non-linearity:** We utilize LeakyReLU to process entropy differences, effectively handling the non-linear relationship between entropy decay and node importance. This prevents vanishing gradients when capturing subtle "entropy decays" in stable systems.
- **Structural Awareness via Element-wise Multiplication:** The final prediction function F relies on the element-wise multiplication of aggregated representations z_i^m and learnable vectors v_i . This multiplication ($z_i^m \odot v_i$) is critical, as it allows the GNN to differentiate between "temporal oscillations" (the signal) and "structural position" (the topological importance), significantly improving accuracy over standard GNN baselines.

4. Empirical Synthesis: SMAP, MSL, and SWaT Dataset Analysis

The framework was validated using NASA's Soil Moisture Active Passive (SMAP) and Mars Science Laboratory (MSL) datasets, alongside the Secure Water Treatment (SWaT) and Water Distribution (WADI) benchmarks.

Quantitative Results

Dataset, Precision, Recall, F1-Score, Top Baseline F1

SMAP, 0.8452, 0.9790, 0.9071, 0.8985 (ATUAD)

MSL, 0.9182, 0.9941, 0.9546, 0.9591 (GDN)

SWaT, 0.9828, 0.6563, 0.7870, 0.8469 (LSTM)

WADI, 0.3029, 0.8196, 0.4423, 0.4403 (ATUAD)

While the GDN baseline shows a slightly higher F1-score on the MSL dataset (0.9591 vs. 0.9546), our entropy-based approach provides superior Precision on SWaT (0.9828) and the best overall F1 balance for WADI. In high-reliability engineering, the goal is often the minimization of false positives in critical infrastructure, where our method excels.

Ablation Study and Reliability Impact

The "strategic proof" of the entropy mechanism is found in the ablation study. When Entropy Centrality was removed, the MSL F1-score plummeted from **0.9546 to 0.8914**. This confirms that entropy-driven weighting is the primary driver of reliability in high-stakes environments like Mars Rover operations, where failing to weight a logically central sensor can lead to system-wide catastrophic failure.

5. Deployment Roadmap and Future Scalability

As CPS dimensions grow, digital hardware—or "overworked silicon"—becomes a bottleneck due to the $O(d^3)$ complexity of matrix inversion and Cholesky sampling. We are currently stuck in a local optimum with traditional GPUs; the path forward requires physics-based hardware such as **Stochastic Processing Units (SPUs)**.

Alternative Scalable Architecture

To overcome the miniaturization limits of inductors (parasitic coupling/crosstalk), we propose a scalable architecture using:

- **RC Unit Cells:** Resistor-capacitor units eliminate the scaling issues of inductors.
- **Resistive Coupling:** An array of resistors connects unit cells such that the stationary distribution remains a normal distribution with a covariance equal to the *inverse* of the conductance matrix.

The Thermodynamic Advantage

The crossover point for "Thermodynamic Advantage" occurs at dimensions $d > 3000$. At this scale, entropy-based physical dynamics offer a transformative edge:

- **Complexity Reduction:** Physical dynamics scale at $O(d^2)$ compared to $O(d^3)$ for digital Cholesky sampling.
- **Energy and Speed:** At $d = 10,000$, thermodynamic hardware provides order-of-magnitude energy savings and speed increases over state-of-the-art GPUs. For the next generation of industrial AI, moving toward entropy-driven monitoring on thermodynamic hardware is no longer a theoretical preference—it is the only scalable path for maintaining the world's most critical cyber-physical infrastructures.